



TradePulse – Complete Tech Stack Recommendation

Version: 1.0

Date: September 2026

Architecture: PWA + Flutter Mobile App + Shared Backend

Executive Summary

TradePulse will use a **modern, scalable, and cost-effective tech stack** optimized for:

- **South African township conditions** (intermittent connectivity, low-end devices)
- **Rapid development** (PWA first, Flutter later)
- **Scalability** (1,000 → 100,000+ users)
- **Security & compliance** (POPIA, financial data)
- **Cost efficiency** (startup-friendly pricing)

1. Frontend Stack

1.1 Progressive Web App (PWA)

Layer	Technology	Version	Why
Framework	Next.js	14+ (App Router)	SSR/SSG, excellent PWA support, Vercel integration
Language	TypeScript	5.3+	Type safety, better DX, fewer bugs
Styling	Tailwind CSS	3.4+	Rapid UI development, mobile-first, small bundle
UI Components	shadcn/ui	Latest	Accessible, customizable, Radix UI based
State Management	Zustand	4.5+	Lightweight, simple, perfect for PWA
Server State	TanStack Query (React Query)	5.0+	Caching, background sync, offline support
Forms	React Hook Form	7.49+	Performance, validation, easy integration
Validation	Zod	3.22+	Schema validation, TypeScript integration
Local Database	Dexie.js (IndexedDB wrapper)	4.0+	Offline storage, sync queue, 500MB+ capacity

Layer	Technology	Version	Why
HTTP Client	Axios	1.6+	Interceptors, error handling, mature
PWA Plugin	next-pwa	5.6+	Service worker generation, caching strategies
Voice Recording	Web Audio API + MediaRecorder	Native	No dependencies, works offline
Speech-to-Text	Web Speech API (optional)	Native	Client-side transcription (fallback to server)
Charts	Recharts	2.12+	Responsive, composable, good for dashboards
PDF Generation	@react-pdf/renderer	3.4+	Client-side PDF for passports
Push Notifications	Firebase Cloud Messaging	Latest	Cross-browser, free tier
Analytics	Google Analytics 4 + Vercel Analytics	Latest	Privacy-focused, performance metrics
Error Tracking	Sentry	7.90+	Real-time error monitoring, source maps
Icons	Lucide React	Latest	Lightweight, consistent, accessible
Testing	Jest + React Testing Library + Playwright	Latest	Unit, integration, E2E testing

Estimated Development Time: 6–8 weeks

Estimated Cost: R300,000–R600,000

1.2 Flutter Mobile App

Layer	Technology	Version	Why
Framework	Flutter	3.24+	Cross-platform (iOS + Android), near-native performance
Language	Dart	3.5+	Fast development, strong typing, Flutter-optimized
State Management	Riverpod	2.4+	Modern, testable, scalable (better than Provider)
Architecture	Clean Architecture	N/A	Separation of concerns, testability, maintainability
Local Database	Isar	3.1+	Fast, offline-first, encryption support, Flutter-native
Alternative DB	Hive	2.2+	Lightweight NoSQL, good for simple data
Secure Storage	flutter_secure_storage	5.1+	Encrypted key-value storage (tokens, PINs)
HTTP Client	Dio	5.3+	Interceptors, file upload, cancellation
Voice Recording	record + audioplayers	Latest	Native audio APIs, background recording
Speech-to-Text	speech_to_text	6.3+	Native platform APIs, offline support

Layer	Technology	Version	Why
Push Notifications	firebase_messaging	14.6+	FCM integration, background notifications
Biometric Auth	local_auth	2.1+	Fingerprint, face ID (Android + iOS)
File Picker	file_picker	6.1+	CSV import, document handling
Image Picker	image_picker	1.0+	Product photos, receipt scanning
PDF Generation	pdf + printing	6.0+	Passport generation, report exports
Charts	fl_chart	0.65+	Beautiful, interactive, Flutter-native
Dependency Injection	GetIt	7.6+	Fast, simple, testable
Analytics	firebase_analytics	10.6+	User behavior, funnels, retention
Crash Reporting	firebase_crashlytics	3.4+	Real-time crash tracking
Deep Linking	app_links	6.2+	WhatsApp deep links, password reset
Testing	flutter_test + integration_test + Patrol	Latest	Unit, widget, integration, E2E

Estimated Development Time: 12–16 weeks

Estimated Cost: R900,000–R1,500,000

2. Backend Stack

2.1 Core Backend

Layer	Technology	Version	Why
Runtime	Node.js	20 LTS	Fast, scalable, huge ecosystem, same language as frontend
Framework	Fastify	4.x	Faster than Express, low overhead, TypeScript support
Alternative	Express	4.x	Mature, huge ecosystem, more plugins
Language	TypeScript	5.3+	Type safety, shared types with frontend
API Style	REST + OpenAPI	3.0	Simple, well-understood, easy documentation
Alternative	GraphQL	N/A	Flexible queries, but more complex (use if needed later)
Authentication	NextAuth.js or Auth0	Latest	OTP, social login, session management
Authorization	CASL or AccessControl	Latest	Role-based access control (RBAC)
Validation	Zod	3.22+	Shared schemas with frontend, type inference
ORM	Prisma	5.10+	Type-safe, auto-migrations, excellent DX
Alternative	Drizzle ORM	Latest	Lightweight, SQL-like, faster than Prisma
Database	PostgreSQL	15+	Relational, ACID-compliant, JSON support
Managed DB	Supabase	Latest	PostgreSQL + auth + realtime + storage (all-in-one)

Layer	Technology	Version	Why
Alternative	Railway PostgreSQL	Latest	Simple, scalable, good pricing
Cache	Redis	7+	Sessions, rate limiting, background jobs
Managed Redis	Upstash	Latest	Serverless Redis, pay-per-request
Queue	BullMQ	5.0+	Redis-based, reliable, TypeScript support
File Storage	AWS S3 or Cloudflare R2	Standard	Voice notes, PDFs, images
CDN	Cloudflare	Latest	Global edge caching, DDoS protection

Estimated Development Time: 8–12 weeks

Estimated Cost: R600,000–R1,000,000

2.2 AI & Machine Learning

Service	Technology	Provider	Cost Estimate
Transaction Parsing	GPT-4o-mini	OpenAI API	\$0.15/1M input tokens, \$0.60/1M output tokens
Alternative	Claude 3 Haiku	Anthropic	\$0.25/1M input, \$1.25/1M output
Speech-to-Text	Whisper API	OpenAI	\$0.006/minute (very affordable)
Alternative	Google Cloud Speech-to-Text	Google	\$0.016/minute (first 500k min/month free)
Structured Output	OpenAI Structured Outputs	OpenAI	Included in API pricing
Fallback Parser	Custom NLP (spaCy)	Self-hosted	Free, but less accurate
Voice Activity Detection	Web Audio API	Native	Free (client-side)

Monthly AI Cost Estimate:

- 1,000 users: R5,000–R15,000/month
- 10,000 users: R20,000–R50,000/month
- 100,000 users: R100,000–R250,000/month

2.3 Background Jobs & Async Processing

Service	Technology	Why
Job Queue	BullMQ + Redis	Reliable, retry logic, priority queues
Scheduled Jobs	node-cron or BullMQ Scheduler	Daily reports, cleanup tasks
Email Sending	Resend or SendGrid	Transactional emails (password reset, notifications)

Service	Technology	Why
SMS/OTP	Twilio or MessageBird	Phone verification, critical alerts
WhatsApp (Optional)	Meta WhatsApp Cloud API	Notifications, low-stock alerts
Webhook Handler	Express middleware	Incoming WhatsApp, payment webhooks
Rate Limiting	express-rate-limit or custom Redis	Prevent abuse, protect API
Idempotency	Custom middleware + Redis	Prevent duplicate transactions

3. Infrastructure & DevOps

3.1 Hosting

Component	Service	Tier	Monthly Cost (ZAR)
PWA Hosting	Vercel	Pro	R400–R2,000
Backend Hosting	Render	Starter → Standard	R200–R1,000
Alternative Backend	Railway	Hobby → Pro	R300–R1,500
Database	Supabase	Pro	R800–R4,000
Alternative DB	Railway PostgreSQL	Standard	R500–R2,500
Redis	Upstash	Pay-as-you-go	R200–R1,000
File Storage	Cloudflare R2	Standard	R100–R500
Alternative Storage	AWS S3	Standard	R200–R1,000
CDN	Cloudflare	Free → Pro	R0–R400
Monitoring	Sentry	Team	R800–R2,500
Alternative Monitoring	Datadog	Starter	R1,500–R5,000
Uptime Monitoring	UptimeRobot or Pingdom	Free → Standard	R0–R400

Total Monthly Infrastructure Cost:

- MVP (1,000 users): R3,000–R8,000/month
- Growth (10,000 users): R10,000–R25,000/month
- Scale (100,000 users): R40,000–R100,000/month

3.2 CI/CD & Version Control

Service	Technology	Why
Version Control	GitHub	Industry standard, free for open source
CI/CD (PWA)	Vercel Automatic Deployments	Zero config, preview deployments

Service	Technology	Why
CI/CD (Backend)	GitHub Actions	Free, integrates with GitHub
CI/CD (Flutter)	GitHub Actions + Fastlane	Automated builds, app store deployment
App Distribution (Beta)	Firebase App Distribution	Easy beta testing, feedback collection
App Store (iOS)	Apple App Store Connect	Required for iOS distribution
App Store (Android)	Google Play Console	Required for Android distribution
Secrets Management	GitHub Secrets + Vercel Environment Variables	Secure, easy to manage
Production Secrets	AWS Secrets Manager or Doppler	Enterprise-grade secret rotation

3.3 Monitoring & Observability

Service	Technology	Purpose
Error Tracking	Sentry	Real-time error monitoring, user context
Performance Monitoring	Vercel Analytics + Datadog	Web vitals, API latency, database queries
Uptime Monitoring	UptimeRobot or Pingdom	Downtime alerts, SLA tracking
Logging	Winston + Datadog Logs	Structured logging, search, alerts
APM	Datadog APM or New Relic	End-to-end tracing, bottleneck detection
Database Monitoring	pg_stat_statements (PostgreSQL)	Query performance, slow query detection
Business Metrics	Mixpanel or Amplitude	User behavior, funnels, retention
Server Metrics	Prometheus + Grafana (self-hosted)	CPU, memory, disk, network monitoring

4. Security Stack

4.1 Application Security

Layer	Technology	Implementation
HTTPS	Let's Encrypt (via Vercel/Cloudflare)	Free SSL, auto-renewal
Authentication	OTP (Twilio/MessageBird) + JWT	Phone verification, 24h token expiry
Authorization	CASL or custom RBAC	Trader, distributor, bank, admin roles
Input Validation	Zod schemas	Server-side + client-side validation
SQL Injection Prevention	Prisma ORM (parameterized queries)	No raw SQL
XSS Prevention	Content Security Policy (CSP)	Strict CSP headers
CSRF Protection	SameSite cookies + CSRF tokens	Prevent cross-site attacks
Rate Limiting	express-rate-limit + Redis	100 req/min per user, 1000 req/min per IP

Layer	Technology	Implementation
DDoS Protection	Cloudflare	Free tier, automatic mitigation
Secrets Management	GitHub Secrets + Environment Variables	No hardcoded secrets
Dependency Scanning	GitHub Dependabot + Snyk	Automated vulnerability detection
Security Headers	Helmet.js	X-Frame-Options, X-Content-Type-Options, etc.
Audit Logging	Custom middleware + PostgreSQL	All actions logged with user ID, timestamp

4.2 Data Protection

Requirement	Implementation
Encryption at Rest	PostgreSQL TDE (Transparent Data Encryption) + S3 server-side encryption
Encryption in Transit	TLS 1.3 for all API communication
Local Storage Encryption	Isar encryption (Flutter) + IndexedDB encryption (PWA)
Secure Token Storage	flutter_secure_storage (Flutter) + HTTP-only cookies (PWA)
POPIA Compliance	Consent management, data minimisation, right to deletion
Data Retention	Automated cleanup jobs (delete old voice notes, logs)
Backup Strategy	Daily automated backups (PostgreSQL) + S3 versioning
Disaster Recovery	Multi-region S3 replication + database point-in-time recovery

5. Third-Party Services

5.1 Communication

Service	Provider	Cost	Use Case
SMS/OTP	Twilio	~R0.50/SMS	Phone verification, critical alerts
Alternative SMS	MessageBird	~R0.40/SMS	Cheaper for SA market
Email	Resend	Free → R200/month	Password reset, notifications
Alternative Email	SendGrid	Free → R300/month	More features, higher limits
WhatsApp (Optional)	Meta Cloud API	R0.08–R1.50/conversation	Notifications, alerts
Push Notifications	Firebase Cloud Messaging	Free	App notifications, reminders

5.2 Payments (Future)

Service	Provider	Cost	Use Case
Payment Gateway	PayFast	3.5% + R7	Trader subscriptions, premium features
Alternative	Yoco	2.95%	Card payments, invoicing
Bank Transfers	Ozow	1.5%	EFT payments, bulk transfers
Mobile Money	M-Pesa (future)	Variable	Rural traders without bank accounts

5.3 Analytics & Business Intelligence

Service	Provider	Cost	Use Case
Product Analytics	Mixpanel	Free → R2,000/month	User behavior, funnels, retention
Alternative	Amplitude	Free → R3,000/month	More advanced analytics
Web Analytics	Google Analytics 4	Free	Traffic, conversions, SEO
Privacy-First	Plausible	R200/month	GDPR-compliant, lightweight
Session Recording	LogRocket or Hotjar	R1,500/month	User session replay, UX insights
A/B Testing	Optimizely or VWO	R5,000+/month	Conversion optimization

6. Development Tools

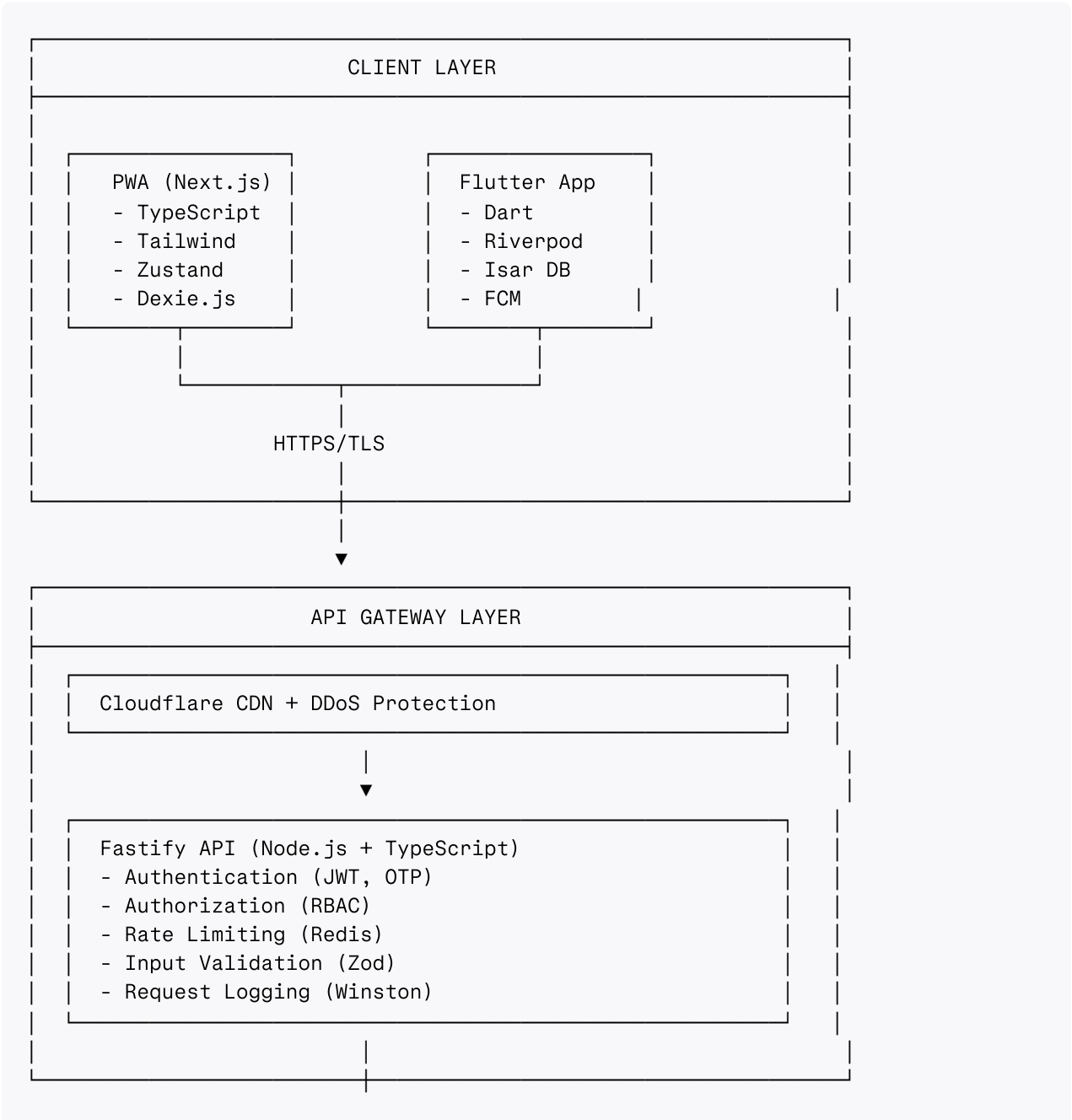
6.1 IDE & Code Quality

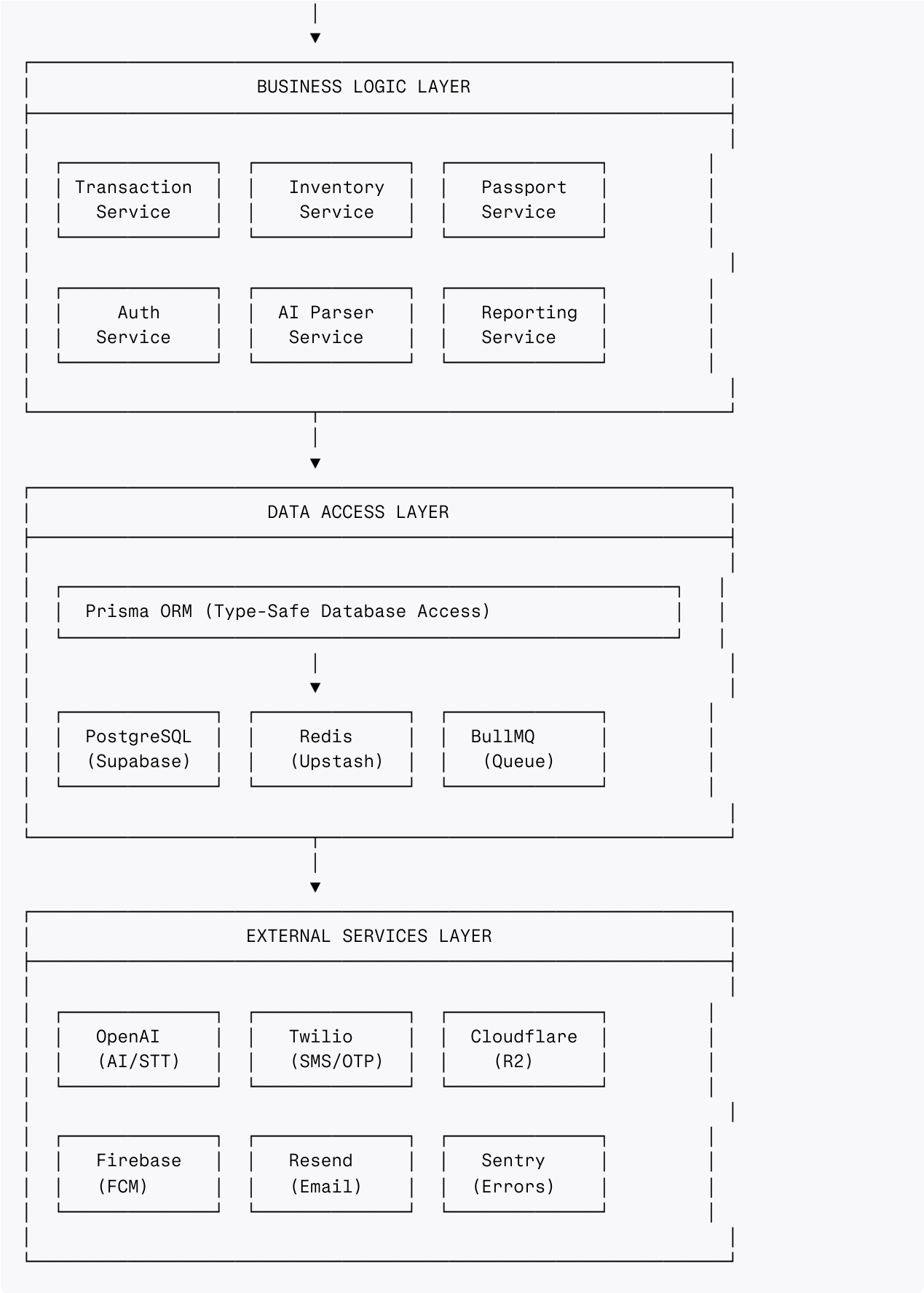
Tool	Purpose	Cost
VS Code	Primary IDE	Free
Android Studio	Flutter development	Free
Xcode	iOS development	Free (macOS only)
Prettier	Code formatting	Free
ESLint	Linting (JavaScript/TypeScript)	Free
Dart Analyzer	Linting (Dart)	Built-in
Husky + lint-staged	Pre-commit hooks	Free
Commitlint	Conventional commits	Free

6.2 Project Management

Tool	Purpose	Cost
GitHub Projects	Issue tracking, Kanban boards	Free
Linear	Faster alternative to GitHub Projects	Free → R200/user/month
Notion	Documentation, wikis, roadmaps	Free → R200/user/month
Figma	UI/UX design, prototyping	Free → R300/user/month
Slack	Team communication	Free → R200/user/month
Discord	Community building	Free

7. Complete Architecture Diagram





8. Cost Summary

8.1 Development Costs (One-Time)

Component	Cost Range (ZAR)
PWA Development (6–8 weeks)	R300,000–R600,000
Flutter App Development (12–16 weeks)	R900,000–R1,500,000
Backend Development (8–12 weeks)	R600,000–R1,000,000
UI/UX Design	R150,000–R300,000
Project Management	R200,000–R400,000
Testing & QA	R150,000–R300,000
Total Development	R2,300,000–R4,100,000

8.2 Monthly Operating Costs

Component	MVP (1k users)	Growth (10k users)	Scale (100k users)
Hosting (Vercel + Render)	R600–R3,000	R2,000–R8,000	R8,000–R20,000
Database (Supabase/Railway)	R800–R4,000	R2,000–R8,000	R8,000–R20,000
Redis (Upstash)	R200–R1,000	R500–R2,000	R2,000–R5,000
Storage (R2/S3)	R100–R500	R500–R2,000	R2,000–R8,000
AI/ML (OpenAI)	R5,000–R15,000	R20,000–R50,000	R100,000–R250,000
SMS/OTP (Twilio)	R2,000–R5,000	R10,000–R25,000	R50,000–R100,000
Monitoring (Sentry + Datadog)	R1,600–R5,000	R3,000–R10,000	R10,000–R25,000
Email (Resend)	R0–R200	R200–R500	R500–R2,000
CDN (Cloudflare)	R0–R400	R400–R800	R800–R2,000
Total Monthly	R10,300–R34,100	R38,600–R106,300	R181,300–R432,000

9. Recommended Stack for Each Phase

Phase 1: MVP (Months 1–3)

Focus: Validate product-market fit with minimal cost

Component	Choice
Frontend	PWA only (Next.js + Tailwind)
Backend	Fastify + PostgreSQL (Supabase)

Component	Choice
AI	OpenAI GPT-4o-mini + Whisper
Hosting	Vercel (PWA) + Supabase (Backend + DB)
SMS	Twilio (pay-as-you-go)
Monitoring	Sentry (free tier) + Vercel Analytics
Budget	R10,000–R20,000/month

Phase 2: Growth (Months 4–12)

Focus: Scale to 10,000 users, add Flutter app

Component	Choice
Frontend	PWA + Flutter app
Backend	Fastify + PostgreSQL (Railway)
AI	OpenAI + custom NLP fallback
Hosting	Vercel Pro + Railway Pro
SMS	MessageBird (volume discount)
Monitoring	Sentry Team + Datadog
Budget	R40,000–R100,000/month

Phase 3: Scale (Months 12+)

Focus: 100,000+ users, enterprise features

Component	Choice
Frontend	PWA + Flutter (iOS + Android)
Backend	Microservices (Node.js + Go)
Database	PostgreSQL (managed) + Read replicas
AI	Fine-tuned models + custom NLP
Hosting	AWS/GCP (multi-region)
SMS	Direct carrier integration
Monitoring	Datadog Enterprise + custom dashboards
Budget	R200,000–R500,000/month

10. Key Decisions & Trade-offs

Why Next.js for PWA?

- ✓ Server-side rendering for SEO
- ✓ Excellent Vercel integration
- ✓ Large ecosystem, easy hiring
- ✗ Larger bundle size than alternatives

Why Flutter for Mobile?

- ✓ Single codebase for iOS + Android
- ✓ Near-native performance
- ✓ Fast development with hot reload
- ✗ Larger app size than native
- ✗ Some platform-specific features require native code

Why Supabase for MVP?

- ✓ All-in-one (PostgreSQL + Auth + Storage)
- ✓ Free tier generous
- ✓ Real-time subscriptions built-in
- ✗ Vendor lock-in
- ✗ Less flexible than self-managed PostgreSQL

Why OpenAI for AI?

- ✓ Best-in-class accuracy
- ✓ Structured outputs feature
- ✓ Easy integration, good documentation
- ✗ Cost at scale
- ✗ Data privacy concerns (use enterprise tier)

Why Vercel for Hosting?

- ✓ Zero-config deployments
- ✓ Automatic HTTPS, CDN, scaling
- ✓ Preview deployments for every PR
- ✗ Can get expensive at scale
- ✗ Less control than self-managed infrastructure

11. Final Recommendation

For MVP (Months 1–6):

- **PWA:** Next.js 14 + Tailwind CSS + shadcn/ui
- **Backend:** Fastify + Supabase (PostgreSQL + Auth + Storage)

- **AI:** OpenAI GPT-4o-mini + Whisper
- **Hosting:** Vercel (PWA) + Supabase (Backend)
- **Monitoring:** Sentry + Vercel Analytics
- **Budget:** R10,000–R20,000/month

For Growth (Months 7–18):

- **Add:** Flutter mobile app
- **Migrate:** Backend to Railway or self-managed PostgreSQL
- **Scale:** AI usage, add custom NLP fallback
- **Budget:** R40,000–R100,000/month

For Scale (Months 19+):

- **Consider:** Microservices architecture
- **Add:** Multi-region deployment
- **Optimize:** AI costs with fine-tuned models
- **Budget:** R200,000–R500,000/month

This stack balances **speed of development**, **cost efficiency**, **scalability**, and **security**—perfect for a startup targeting the South African township market with plans for regional expansion.